Problem 5

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Problem 1 *Differentiate each function (with respect to x)*

(a) $\cos\left(\sqrt{x+7}\right)$

Explanation.

$$\begin{aligned}\frac{d}{dx}\left(\cos\left(\sqrt{x+7}\right)\right) &= -\sin\left(\sqrt{x+7}\right)\cdot\frac{1}{2}(x+7)^{\frac{-1}{2}}(1)\\ &= \frac{-\sin\left(\sqrt{x+7}\right)}{2\sqrt{x+7}}\end{aligned}$$

(b) $\sqrt{\cos(x)+7}$

Explanation.

$$\begin{aligned}\frac{d}{dx}\left(\sqrt{\cos(x)+7}\right) &= \frac{1}{2}\left(\cos(x)+7\right)^{\frac{-1}{2}}\left(-\sin(x)\right)\\ &= \frac{-\sin(x)}{2\sqrt{\cos(x)+7}}\end{aligned}$$

(c) $\sqrt{\cos(x)}+7$

Explanation.

$$\begin{aligned}\frac{d}{dx}\left(\sqrt{\cos(x)}+7\right) &= \frac{1}{2}\left(\cos(x)\right)^{\frac{-1}{2}}\left(-\sin(x)\right)+0\\ &= \frac{-\sin(x)}{2\sqrt{\cos(x)}}\end{aligned}$$

(d) $\cos\left(\sqrt{x}+7\right)$

Explanation.

$$\begin{aligned}\frac{d}{dx}\left(\cos\left(\sqrt{x}+7\right)\right) &= -\sin\left(\sqrt{x}+7\right)\cdot\frac{1}{2}x^{\frac{-1}{2}}\\ &= \frac{-\sin\left(\sqrt{x}+7\right)}{2\sqrt{x}}\end{aligned}$$

(e) $\cos(x)\cdot\left(\sqrt{x}+7\right)$

Explanation.

$$\begin{aligned}\frac{d}{dx}\left(\cos(x)\cdot\left(\sqrt{x}+7\right)\right) &= -\sin(x)\cdot\left(\sqrt{x}+7\right)+\cos(x)\cdot\frac{1}{2}x^{\frac{-1}{2}}\\ &= -\sin(x)\cdot\left(\sqrt{x}+7\right)+\frac{\cos(x)}{2\sqrt{x}}\end{aligned}$$